

1 Natural Language State Description

The RoboRacer platform operates in a partially observable (yet structured) environment where the vehicle derives its own pose from the on-board LIDAR and depth-sensing camera modules alongside velocity estimation from individual wheel RPM sensors and targeted throttle settings. Paired with a memory of recent estimations from previous discrete timesteps, the system is remarkably effective in deriving critical positional information such as global yaw, proximity to walls and other vehicles, and even advanced inferences like wheel slip detection and attempted vs actual steering angle achieved.

The LIDAR module provides a set of measurements that are returned to the central data collection subsystem as a dictionary of angles (relative to the car's forward heading) and distances to where those rays intersect an obstacle, down to roughly millimeter-level precision. The depth-sensing camera provides similar functionality, but can be extended with computer vision algorithms to track specific points of interest, like other competing vehicles, over time. This functionality is likely outside the scope of this project and our state vector will primarily be composed of the returned LIDAR distance rays.

Additional empirical measurements, such as tracked yaw and lateral acceleration from the IMU, round out the state. Other systems information like battery voltage and discharge rate may also be included due to their effects on maximum instantaneous torque which may have implications for the optimal inputs chosen by the MPC process.

2 Mathematical State Description

Adapted from main research notes

Note that the observation vector is collapsed into the state vector.

$$\mathbf{x}_k = \left[\mathbf{l}_k^\top, \mathbf{d}_k^\top, \omega_k^{FL}, \omega_k^{FR}, \omega_k^{RL}, \omega_k^{RR}, \delta_k^a, \delta_k^d, \mathbf{u}_{k-1}^\top, \psi_k, a_{y,k}, V_{\text{bat},k} \right]^\top$$

- **LIDAR range array** $\mathbf{l}_k$: A vector of distance measurements spanning the forward 180 degree field of view, where $l_k^{(i)}$ is the range return for the i -th ray at angle θ_i .
- **Depth image** $\mathbf{d}_k$: A matrix of per-pixel depth values from the forward-facing depth camera, where $d_k^{(r,c)}$ gives the metric depth at row r , column c . This may or may not be employed in the actual algorithm. If it is, it will primarily be used as a reinforcement of the pose estimation based on the LIDAR.
- **Wheel angular velocities** $\omega_k^{FL}, \omega_k^{FR}, \omega_k^{RL}, \omega_k^{RR}$: RPM readings from the front-left, front-right, rear-left, and rear-right wheel encoders.
- **Actual steering angle** δ_k^a : The measured servo position representing the true front-wheel steering angle. (This may need to be either estimated from empirically measured turning rates or implemented in hardware, I need to read the servo datasheet)
- **Desired steering angle** δ_k^d : The most recently commanded steering setpoint.

- **Previous control input** $\mathbf{u}_{k-1} = [\tau_{k-1}, \delta_{k-1}^{\text{cmd}}]^\top$: The throttle command τ and steering command δ^{cmd} applied at the preceding time step.
- **Yaw rate** ψ_k : Angular velocity about the vertical axis, measured by the IMU at timestep k .
- **Lateral acceleration** $a_{y,k} \in \mathbb{R}$: Sideways acceleration measured by the IMU at timestep k .
- **Battery voltage** $V_{\text{bat},k}$: Pack voltage measured by the ESC at timestep k .

3 Control Input Vector (or Action Vector)

$$\mathbf{u}_k^\top = [\tau_k, \delta_k^{\text{cmd}}]$$

Where τ is the throttle command and δ is the steering command.

4 State Transition Model

The state evolves in the internal model by a smooth set of linear equations based upon the kinematic bicycle model of equations (*adapted from main research notes*).

$$\begin{aligned}\mathbf{x} &= [x \quad y \quad \psi \quad v]^\top, \\ \mathbf{u} &= [\delta \quad a]^\top\end{aligned}$$

where (x, y) is the position of the vehicle reference point (taken here at the center of mass), ψ is the heading angle measured from the positive x -axis, v is the longitudinal speed, δ is the front-wheel steering angle, and a is the longitudinal acceleration.

Slip Angle

Because the reference point is the center of mass rather than the rear axle, a geometric slip angle β appears:

$$\beta = \arctan\left(\frac{l_r \tan \delta}{L}\right),$$

where l_r is the distance along the vehicle centerline from the rear axle to the center of mass and L is the total wheelbase.

Equations of Motion (computed over a single timestep)

$$\begin{aligned}x &= v \cos(\psi + \beta), \\ y &= v \sin(\psi + \beta), \\ \psi &= \frac{v \cos \beta}{L} \tan \delta, \\ v &= a\end{aligned}$$